

SIMATS ENGINEERING
CO1 - ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0405

Name of the Student	Guggilla Venkatasai
Reg. No	192425388

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 1 Hour

Marks: 50

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Answer:

Step 1: Count the outcomes

- Number of Heads (H) = 7
- Number of Tails (T) = 3
- Total tosses (n) = 10

Step 2: MLE Formula

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Step 3: Substitute the values

$$\hat{p} = \frac{7}{10} = 0.7$$

Final Answer

$$\boxed{\hat{p} = 0.7}$$

Interpretation:

The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is 0.7 (70%). This means the coin is estimated to produce Heads with a probability of 70% based on the observed data.

2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Answer:

Step 1: Given Data

- Number of students who passed = **42**
- Total number of students = **50**

Step 2: MLE Formula

For a Bernoulli distribution, the Maximum Likelihood Estimator (MLE) of the probability of success is:

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Observations}}$$

Step 3: Substitute the values

$$\hat{p} = \frac{42}{50} = 0.84$$

Final Answer

$$\boxed{\hat{p} = 0.84}$$

Interpretation

The Maximum Likelihood Estimate (MLE) of the probability that a student passes the exam is 0.84 (84%). This means that, based on the observed data, a randomly selected student has an estimated 84% probability of passing the exam.

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.

Answer:

Step 1: Given Data

- Total number of bulbs = **100**
- Number of defective bulbs = **8**

Step 2: MLE Formula

For a Bernoulli distribution, the Maximum Likelihood Estimator (MLE) of the probability of success is:

$$\hat{p} = \frac{\text{Number of Defective Bulbs}}{\text{Total Number of Bulbs}}$$

Step 3: Substitute the values

$$\hat{p} = \frac{8}{100} = 0.08$$

Final Answer

$$\boxed{\hat{p} = 0.08}$$

Interpretation

The Maximum Likelihood Estimate (MLE) of the probability that a light bulb is defective is 0.08 (8%). This means that, based on the observed sample, a randomly selected bulb has an estimated 8% probability of being defective.

4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.

Answer:**Step 1: Given Data**

- Total number of rolls = **60**
- Number of times 6 appears = **18**

Step 2: MLE Formula

For a Bernoulli distribution, the Maximum Likelihood Estimator (MLE) of the probability of success is:

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Observations}}$$

Step 3: Substitute the values

$$\hat{p} = \frac{18}{60} = 0.30$$

Final Answer

$$\boxed{\hat{p} = 0.30}$$

Interpretation

The Maximum Likelihood Estimate (MLE) of the probability of rolling a 6 is 0.30 (30%). This means that, based on the observed data, the biased die has an estimated 30% probability of producing the number 6.

5. Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Solution

Step 1: Calculate the Predicted Output

Formula:

$$y = w \times x$$

Substitute the given values:

$$y = 2 \times 4 = 8$$

Predicted Output = 8

Step 2: Calculate the Error

Formula:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Substitute the values:

$$\text{Error} = 10 - 8 = 2$$

Error = 2

Step 3: Update the Weight

Gradient Descent Update Rule:

$$w_{\text{new}} = w - \eta \times \text{Gradient}$$

Where:

- $w = 2$
- Learning rate (η) = 0.01
- Gradient = 4

Substitute the values:

$$\begin{aligned} w_{\text{new}} &= 2 - (0.01 \times 4) \\ w_{\text{new}} &= 2 - 0.04 = 1.96 \end{aligned}$$

Updated Weight = 1.96

Code of Conduct:

I **Guggilla Venkatasai(192425388)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Name of the student:

Guggilla Venkatasai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation